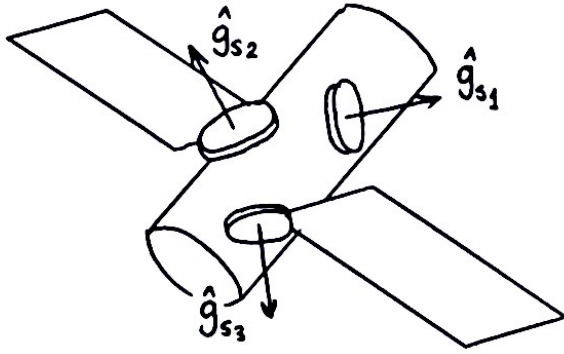


Consider the rigid spacecraft with 3 Reaction Wheels (RWs) shown:



The wheels have generic spin axis orientation $\hat{g}_{si}$, which are fixed relative to the body B . Each RW is identical in build and has the same spin axis inertia I_w , while the time varying RW spin rate relative to the body is given through $\Omega_i(t)$.

Each body-fixed reaction wheel frame is defined through: $G_i \equiv \{ \hat{g}_{si}, \hat{g}_{ti}, \hat{g}_{gi} \}$

The inertia tensor $[I]$ contains the inertia of the spacecraft and the stationary wheels.

(Assume the spacecraft is experiencing a general disturbance torque $\vec{\tau}_c$ about the CM.)

A) The angular momentum of this spacecraft about its center of mass is expressed as:

$$\vec{H}_c = [I]\vec{\omega} + I_w \sum_{i=1}^3 \Omega_i \hat{g}_{si} \quad \text{Derive the rotational equations of motion for this S/C.}$$

(Recall that $\omega_{si} = \vec{\omega} \cdot \hat{g}_{si}$)

The rate of change of the total angular momentum $\vec{H}_c$ with respect to an inertial frame (N) is equal to the net torque $\vec{\tau}_c$ acting on the system's center of mass: $\dot{\vec{H}}_c|_N = \vec{\tau}_c$

Since the spacecraft is rotating with angular velocity $\vec{\omega}$, and our parameters are defined in the body-fixed frame B , we must relate the inertial derivative to the body-frame derivative using the Transport Theorem: $\dot{\vec{H}}_c|_N = \dot{\vec{H}}_c|_B + \vec{\omega} \times \vec{H}_c$

The total angular momentum provided is:

$$\vec{H}_c = \underbrace{[I]\vec{\omega}}_{\text{Spacecraft Momentum}} + \underbrace{I_w \sum_{i=1}^3 \Omega_i \hat{g}_{si}}_{\text{Relative wheel Momentum}}$$

We calculate the time derivative as seen by an observer of the spacecraft. In frame B , the inertia tensor $[I]$ and the wheel spin axes $\hat{g}_{si}$ are constant.

$$\dot{\vec{H}}_c|_B = [I]\dot{\vec{\omega}} + I_w \sum_{i=1}^3 \dot{\Omega}_i \hat{g}_{si} \quad \leftarrow \quad \dot{\vec{\omega}}|_N = \dot{\vec{\omega}}|_B + \vec{\omega} \times \vec{\omega} = \dot{\vec{\omega}}|_B \quad (\text{Transport Theorem})$$

Substituting: $\vec{\tau}_c = \dot{\vec{H}}_c|_N = \underbrace{[I]\dot{\vec{\omega}} + I_w \sum_{i=1}^3 \dot{\Omega}_i \hat{g}_{si}}_{\dot{\vec{H}}_c|_B} + \underbrace{\vec{\omega} \times ([I]\vec{\omega} + I_w \sum_{i=1}^3 \Omega_i \hat{g}_{si})}_{\vec{\omega} \times \vec{H}_c}$

If we define the wheels angular momentum as $\vec{h}_{rw} = I_w \sum_{i=1}^3 \Omega_i \hat{g}_{si}$

$$\vec{\tau}_c = [I]\dot{\vec{\omega}} + \vec{h}_{rw} + \vec{\omega} \times ([I]\vec{\omega} + \vec{h}_{rw})$$

Finally, we can rearrange the terms to isolate the angular acceleration:

$$[I]\dot{\vec{\omega}} = -\vec{\omega} \times ([I]\vec{\omega} + \vec{h}_{rw}) - \dot{\vec{h}}_{rw} + \vec{\tau}_c$$

Where: $\vec{h}_{rw} \equiv$ Relative angular momentum of the wheels.

NOTE: To map ω into the RW frame, we define $\vec{\omega} = \omega_{s1} \hat{g}_{s1} + \omega_{t1} \hat{g}_{t1} + \omega_{g1} \hat{g}_{g1}$. Therefore, the cross product results in components in the transverse and gimbal axes.

B) Assume the RWs have a particular configuration where $\hat{g}_{si} = \hat{b}_i$, and $\hat{b}_i$ are principal axes of the body B. Refine the general equations of motion in A) for this scenario.
 (You are looking for 3 scalar differential equations for ω_i)

Following the problem statement, the reaction wheel (RW) spin axes are aligned with the spacecraft's principal axes ($\hat{g}_{si} = \hat{b}_i$)

This implies that the inertia tensor $[I]$ is DIAGONAL. $[I] = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$

The vectors are projected onto the principal axes.

$$\begin{cases} \text{Spacecraft Angular Velocity} & \vec{\omega} = \omega_1 \hat{b}_1 + \omega_2 \hat{b}_2 + \omega_3 \hat{b}_3 \\ \text{Relative RW Angular Momentum} & \vec{h}_{RW} = I_w \Omega_1 \hat{b}_1 + I_w \Omega_2 \hat{b}_2 + I_w \Omega_3 \hat{b}_3 \end{cases}$$

Total Angular Momentum:

$$\vec{H}_c = [I] \vec{\omega} + \vec{h}_{RW} = (I_1 \omega_1 + I_w \Omega_1) \hat{b}_1 + (I_2 \omega_2 + I_w \Omega_2) \hat{b}_2 + (I_3 \omega_3 + I_w \Omega_3) \hat{b}_3$$

From A), the vector equation for motion is: $\vec{\tau}_c = [I] \dot{\vec{\omega}} + \vec{h}_{RW} + \vec{\omega} \times \vec{H}_c$

Where $\vec{\tau}_c = \tau_1 \hat{b}_1 + \tau_2 \hat{b}_2 + \tau_3 \hat{b}_3$ is the external disturbance torque.

To find the scalar equations, we evaluate the cross product $\vec{\omega} \times \vec{H}_c$ and equate the components for each axis:

$$\vec{\omega} \times \vec{H}_c = \begin{bmatrix} \hat{b}_1 & \hat{b}_2 & \hat{b}_3 \\ \omega_1 & \omega_2 & \omega_3 \\ H_1 & H_2 & H_3 \end{bmatrix} = \omega_2 H_3 \hat{b}_1 + \omega_1 H_2 \hat{b}_3 + \omega_3 H_1 \hat{b}_2 - (\omega_2 H_1 \hat{b}_3 + \omega_3 H_2 \hat{b}_1 + \omega_1 H_3 \hat{b}_2) = (I) \hat{b}_1 + (II) \hat{b}_2 + (III) \hat{b}_3$$

$$\text{Axis by axis: } \begin{cases} \text{I: } \omega_2 H_3 - \omega_3 H_2 = \omega_2 (I_3 \omega_3 + I_w \Omega_3) - \omega_3 (I_2 \omega_2 + I_w \Omega_2) \\ \text{II: } \omega_3 H_1 - \omega_1 H_3 = \omega_3 (I_1 \omega_1 + I_w \Omega_1) - \omega_1 (I_3 \omega_3 + I_w \Omega_3) \\ \text{III: } \omega_1 H_2 - \omega_2 H_1 = \omega_1 (I_2 \omega_2 + I_w \Omega_2) - \omega_2 (I_1 \omega_1 + I_w \Omega_1) \end{cases}$$

By summing the results, we obtain:

$$\begin{cases} \tau_1 = I_1 \dot{\omega}_1 + I_w \dot{\Omega}_1 + \omega_2 (I_3 \omega_3 + I_w \Omega_3) - \omega_3 (I_2 \omega_2 + I_w \Omega_2) \\ \tau_2 = I_2 \dot{\omega}_2 + I_w \dot{\Omega}_2 + \omega_3 (I_1 \omega_1 + I_w \Omega_1) - \omega_1 (I_3 \omega_3 + I_w \Omega_3) \\ \tau_3 = I_3 \dot{\omega}_3 + I_w \dot{\Omega}_3 + \omega_1 (I_2 \omega_2 + I_w \Omega_2) - \omega_2 (I_1 \omega_1 + I_w \Omega_1) \end{cases}$$

Expanding and grouping terms:

$$\begin{cases} \tau_1 = I_1 \dot{\omega}_1 + I_w \dot{\Omega}_1 + (I_3 - I_2) \omega_2 \omega_3 + I_w (\omega_2 \Omega_3 - \omega_3 \Omega_2) \\ \tau_2 = I_2 \dot{\omega}_2 + I_w \dot{\Omega}_2 + (I_1 - I_3) \omega_3 \omega_1 + I_w (\omega_3 \Omega_1 - \omega_1 \Omega_3) \\ \tau_3 = I_3 \dot{\omega}_3 + I_w \dot{\Omega}_3 + (I_2 - I_1) \omega_1 \omega_2 + I_w (\omega_1 \Omega_2 - \omega_2 \Omega_1) \end{cases}$$

With: $\vec{\tau}_c = \tau_1 \hat{b}_1 + \tau_2 \hat{b}_2 + \tau_3 \hat{b}_3$

Finally:

$$\begin{cases} \dot{\omega}_1 = (1/I_1) \cdot [\tau_1 - (I_3 - I_2) \omega_2 \omega_3 - I_w \dot{\Omega}_1 - I_w (\omega_2 \Omega_3 - \omega_3 \Omega_2)] \\ \dot{\omega}_2 = (1/I_2) \cdot [\tau_2 - (I_1 - I_3) \omega_3 \omega_1 - I_w \dot{\Omega}_2 - I_w (\omega_3 \Omega_1 - \omega_1 \Omega_3)] \\ \dot{\omega}_3 = (1/I_3) \cdot [\tau_3 - (I_2 - I_1) \omega_1 \omega_2 - I_w \dot{\Omega}_3 - I_w (\omega_1 \Omega_2 - \omega_2 \Omega_1)] \end{cases}$$